Topic 8: The Periodic Table - Answers

IGCSE Chemistry 0620 Paper 4 topical answer booklet.

Subtopic: Groups and trends in the Periodic Table

Questions in this answer PDF: 10

Use this with the matching topical question PDFs. The answer points are organised by the same source labels.

Short explanation notes are added for harder calculation, electrolysis, equilibrium, rate, salts, organic, metals and practical questions.

Periodic table reference pages are not included to save printing paper.

2020 Feb/March Paper 42 Question 4

Main topic: Topic 8: The Periodic Table | Marks: 12 | Answer source: 0620_m20_ms_42.pdf

Extra topic(s): Topic 7: Acids, bases and salts, Topic 9: Metals, Topic 6: Chemical reactions

Answer points

4(a)

- Haber (process) (1)
- ammonia (1)

4(b)(i)

- green

4(b)(ii)

- Fe2+(aq) + 2OH-(aq) -> Fe(OH)2(s)
- Fe(OH)2 (as only product) (1)
- Fe2+ and 2OH- (as reactants) (1)
- state symbols (1)

4(c)(i)

- oxidising agent

4(c)(ii)

- presence of an acid

4(c)(iii)

- lose an electron

4(c)(iv)

- colourless

4(d)

- 3+
- 3+

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2020 Oct/Nov Paper 43 Question 5

Main topic: Topic 8: The Periodic Table | Marks: 16 | Answer source: 0620_w20_ms_43.pdf

Extra topic(s): Topic 4: Electrochemistry

Answer points

5(a)

- become more reactive down the group opposite reasoning also allowed (1)

5(b)(i)

- one mark each for any two of:

- -

- floats

- -

- dissolves / disappears / melts

- -

- moves

- -

- bubbles / fizzes / effervesces

- -

- lilac flame

5(b)(ii)

- $2K + 2H_2O \rightarrow 2KOH + H_2$
- all formulae (1)
- equation fully correct (1)

5(c)(i)

- $Cl_2 + 2KI \rightarrow 2KCl + I_2$
- OR $Cl_2 + 2I^- \rightarrow 2Cl^- + I_2$
- all formulae (1)
- equation fully correct (1)

5(c)(ii)

- brown / black

5(d)(i)

- breakdown by (the passage of) electricity (1)
- of an ionic compound in molten or aqueous (state) (1)

5(d)(ii)

- heat until it melts / heat to or above melting point

5(d)(iii)

- $Na^+ + e \rightarrow Na$

5(e)(i)

- one mark for each of any two from:
- -
- (chromium has) high melting point opposite reasoning also allowed
- -
- (chromium forms) coloured ions / coloured compounds opposite reasoning also allowed
- -
- (chromium has) variable valency / variable oxidation state / variable oxidation number opposite reasoning also allowed
- -
- catalytic behaviour opposite reasoning also allowed
- opposite reasoning also allowed ALLOW group 1 or sodium if stated
- -
- no colour or white or colourless ions or compounds
- -
- fixed valency / +1 charge only or one oxidation state / forms one chloride
- -
- low melting point
- -
- doesn't behave as a catalyst

5(e)(ii)

- one mark for each of any two from:
- -
- (chromium / sodium) conducts electricity
- -
- (chromium / sodium) compounds are soluble (in water)
- -
- (chromium / sodium) form hydrated salts / form hydrated compounds

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2021 Oct/Nov Paper 42 Question 5

Main topic: Topic 8: The Periodic Table | Marks: 22 | Answer source: 0620_w21_ms_42.pdf

Answer points

5(a)

- metallic (1)
- (lattice of) positive ions (1)
- sea of / delocalised / mobile electrons (1)
- attraction between positive ions and electrons (1)

5(b)(i)
- malleability / malleable

5(b)(ii)
- (particles) slide (over each other)

5(b)(iii)
- unreactive

5(c)
- high(er) density
- high(er) melting points

5(d)(i)
- Na - yellow (1)
- K - lilac (1)
- colourless solution (1)
- clear idea of increased reactivity in both reactions (1)

5(d)(ii)
- hydrogen

5(d)(iii)
- potassium hydroxide

5(d)(iv)
- any number in the range 7< pH ■ 14

5(d)(v)
- 4Na + O2 -> 2Na2O
- Na2O (1) correct equation (1)

5(e)(i)
- N2 + 3H2 -> 2NH3
- species (1) correct equation (1)

5(e)(ii)
- 450 °C (1)
- 200 atm (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2022 May/June Paper 41 Question 7

Main topic: Topic 8: The Periodic Table | Marks: 11 | Answer source: 0620_s22_ms_41.pdf

Answer points

7(a)(i)
- from left to right
- caesium -> rubidium-> potassium ->sodium -> lithium

7(a)(ii)
- caesium hydroxide

7(b)
- Group I element is less strong / not strong
- opposite reasoning also allowed
- OR
- Group I element has low(er) density opposite reasoning also allowed
- OR
- Group I element is soft(er) opposite reasoning also allowed

7(c)
- solid

7(d)(i)
- colourless (1)
- orange / brown / yellow (1)

7(d)(ii)
- Br - (1)
- loses electron(s) (1)

7(e)

- 432(1)
- 436(1)
- - 4(1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2022 May/June Paper 43 Question 6

Main topic: Topic 8: The Periodic Table | Marks: 16 | Answer source: 0620_s22_ms_43.pdf

Answer points

6(a)(i)

- Group 1 metals do not show catalytic behaviour
- Group 1 have fixed oxidation states

6(a)(ii)

- any 2 observations from:
- ■
- moves / floats
- ■
- dissolves / disappears
- ■
- bubbles / effervescence / fizzes
- ■
- lilac flame
- ■
- explodes
- ■
- melts / forms a spherical shape
- 2K(s) + 2H2O(l) ->2KOH(aq) + H2(g)
- KOH or H2 product (1) in equation
- fully correct equation (1)
- state symbols (1)

6(b)

- transition elements have high(er) melting point (1)
- transition elements have high(er) density (1)

6(c)(i)

- colourless (1)
- brown (1)

6(c)(ii)

- redox

6(c)(iii)

- Br2 (1)
- (bromine) is reduced (1)

6(d)

- two ticks in first row (1)
- three crosses in the other three boxes (1)

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2023 Feb/March Paper 42 Question 4

Main topic: Topic 8: The Periodic Table | Marks: 18 | Answer source: 0620_m23_ms_42.pdf
Extra topic(s): Topic 3: Stoichiometry, Topic 9: Metals

Answer points

4(a)(i)

- zinc

4(a)(ii)

- alloy

4(b)(i)
- ductility

4(b)(ii)
- electrons

4(c)(i)
- high melting point

4(c)(ii)
- (act as) catalysts

4(d)(i)
- water(s) of crystallisation

4(d)(ii)
- blue

4(d)(iii)
- $CuSO_4$■$5H_2O$
- $CuSO_4$ (1)
- ■$5H_2O$ (1)

4(e)(i)
- basic

4(e)(ii)
- the oxidation number of copper is +2

4(e)(iii)
- Mr of $Cu(NO_3)_2$ = 188 (1)
- = 0.0200 -> = 0.0200 -> 188 = 3.76 g (1)
- February/March 2023

4(e)(iv)
- moles of gas formed = 0.0200 -> 5 / 2 = 0.05(00) (1)
- volume = -> 24.0 = 0.05(00) -> 24.0 = 1.2(0) (1)

4(e)(v)
- $2Al + 3CuO$ -> $Al_2O_3 + 3Cu$
- correct products (1)
- rest of equation correct (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2024 Feb/March Paper 42 Question 2

Main topic: Topic 8: The Periodic Table | Marks: 18 | Answer source: 0620_m24_ms_42.pdf
Extra topic(s): Topic 2: Atoms, elements and compounds

Answer points

2(a)
- fluorine

2(b)
- red-brown AND liquid

2(c)
- Ts
- 7

2(d)(i)
- pair of electrons
- electron(s) shared between two atoms

2(d)(ii)
- iodide / astatide / tenesside

2(d)(iii)
- bromine is more reactive than iodine / astatine / tenessine

2(e)
- cobalt(II) chloride
- anhydrous

2(f)
- 8 crosses in third shell of Ca
- 7 dots and 1 cross in third shell of both Cl
- '2+' charge on Ca on correct answer line AND '-' charge on both Cl ions on correct answer line
- February/March 2024

2(g)(i)
- lead(II) nitrate

2(g)(ii)
- Pb2+(aq) + 2Cl -(aq) -> PbCl2(s)
- PbCl2 as only product
- Pb2+ + 2Cl - as only reactants
- correct state symbols

2(g)(iii)
- silver chloride

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2025 Feb/March Paper 42 Question 3

Main topic: Topic 8: The Periodic Table | Marks: 13 | Answer source: 0620_m25_ms_42.pdf

Answer points

3(a)
- VII

3(b)

3(c)(i)
- tennessine / Ts / Ts2

3(c)(ii)
- fluorine / F / F2

3(d)
- halide(s)

3(e)

3(f)
- krypton

3(g)(i)
- Cl 2 + 2I- -> 2Cl - + I2
- Cl-
- correct ionic equation

3(g)(ii)
- I- loses e -
- Cl2 gains e -

3(h)
- grey-black
- solid
- February/March 2025

Easy explanation / reminder

- *Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.*

2025 May/June Paper 41 Question 6

Main topic: Topic 8: The Periodic Table | Marks: 12 | Answer source: 0620_s25_ms_41.pdf

Extra topic(s): Topic 5: Chemical energetics

Answer points

6(a)
- halogen(s)

6(b)
- fluorine / F2

6(c)
- solid
- grey-black

6(d)(i)
- $Br_2 + 2I^- \rightarrow 2Br^- + I_2$
- I^- as a reactant and Br^- as a product
- equation fully correct

6(d)(ii)
- 343
- 350
- -7

6(e)(i)
- any two from:-
- -
- solid dissolves / solid disappears
- -
- bubbling / effervescence / fizzing
- -
- melts / forms a ball
- -
- floats
- -
- moves

6(e)(ii)
- blue

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

- Energy tip: exothermic releases energy and products are lower. Endothermic absorbs energy and products are higher.

2025 May/June Paper 43 Question 6

Main topic: Topic 8: The Periodic Table | Marks: 15 | Answer source: 0620_s25_ms_43.pdf

Answer points

6(a)
- alkali metals

6(b)
- lithium

6(c)(i)
- Any two:
- -
- solid dissolves / disappears
- -
- fizzing/ bubbles / effervescence
- -
- solid floats
- -
- solid moves (on surface)

6(c)(ii)
- $2Li + 2H_2O \rightarrow 2LiOH + H_2$
- LiOH as product
- correct equation

6(d)
- softer
- lower density

6(e)(i)
- gas
- pale yellow-green

6(e)(ii)
- $Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$
- Br^- as a reactant and Cl^- as a product

- correct equation

6(e)(iii)

- energy required in breaking bonds
- = 150 + 242 = (+) 392
- energy released when bonds are formed
- = 2 -> 218 = (+) 436
- enthalpy change, ΔH
- = 392 - 436 = - 44

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

- Energy tip: exothermic releases energy and products are lower. Endothermic absorbs energy and products are higher.